

Physics-Informed vs Black-Box Learning for Robot Inverse Dynamics: A Comparative Study of DeLAN and FFNN with Normalized Loss Balancing

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Abstract—This study presents a comparative analysis between Feedforward Neural Networks (FFNN) and Deep Lagrangian Networks (DeLAN) for identifying the inverse dynamics of robot manipulators, specifically the Franka Emika PANDA. To address significant torque scaling disparities between proximal and distal joints, we propose a per-joint Normalized Mean Squared Error (nMSE) loss function. This ensures balanced learning across all degrees of freedom (nDOF) by scaling errors relative to each joint’s variance. Furthermore, we investigate the impact of trigonometric input expansion as a structural prior to improve the efficiency of black-box models. Experimental results demonstrate that while deeper FFNNs improve predictive capacity for distal joints, they often violate fundamental physical laws and struggle with out-of-distribution (OOD) data. In contrast, DeLAN ensures physical consistency and energy conservation by embedding the Lagrangian framework into the architecture, yielding superior OOD robustness. However, DeLAN’s reliance on higher-order derivatives increases computational complexity and training time compared to standard FFNNs. This work highlights the critical trade-offs between architectural flexibility and physics-informed inductive biases in robotic regression tasks.

I. INTRODUCTION

The accurate identification of robot manipulator dynamics is fundamental for developing advanced control algorithms and ensuring safe human-robot interaction. Traditionally, inverse dynamics, which calculates the joint torques required to achieve a specific trajectory, is addressed through analytical models based on the Euler-Lagrange equations. However, these models often struggle to account for complex, non linear phenomena such as joint friction, cable dynamics, and manufacturing tolerances. In this context, Deep Learning techniques offer a compelling alternative, enabling the learning of dynamic relationships directly from experimental data without requiring explicit modeling of every physical component.

Despite the success of Feedforward Neural Networks (FFNNs) in non-linear regression, their application to robotics presents significant challenges. A critical issue lies in the torque scaling disparity across different joints: in serial manipulators, proximal joints bear significantly higher loads than distal ones. Using a standard Mean Squared Error (MSE) loss function tends to prioritize joints with larger torque

magnitudes, leading to unacceptable relative errors in the smaller distal joints.

A pivotal aspect of this research is the comparison between purely data-driven and physics-informed architectures.

- **FFNN (Feedforward Neural Networks):** These are chosen for their relative ease of implementation. They are highly effective at capturing complex, non-linear mappings without requiring prior knowledge of the system’s structure. However, they operate as black boxes, often violating fundamental physical laws and showing poor performance when acting beyond the training distribution.
- **DeLAN (Deep Lagrangian Networks):** These are employed to bridge the gap between deep learning and classical mechanics. By enforcing a Lagrangian structure within the network, DeLAN ensures that the learned dynamics are physically consistent.

In this work, we present a comparative study between standard FFNNs and DeLANs for the inverse dynamics identification of 7-DOF manipulators. We propose the adoption of a per-joint Normalized Mean Squared Error (nMSE) loss function, demonstrating how it balances the learning process across joints with different scales. Our results provide insights into how physics-informed biases and loss-function normalization impact accuracy and OOD robustness.

A. Contributions

The primary contributions of this report are summarized as follows:

- A comprehensive comparison between purely data-driven (FFNN) and physics-informed (DeLAN) architectures for robotic regression tasks.
- A critical analysis of the nMSE loss function’s effectiveness as a metric to train the networks compared to traditional MSE.
- An evaluation of model generalization capabilities on out-of-distribution datasets and performance degradation as nDOF increases.

B. Outline

This report is structured as follows. Section II reviews the state of the art in robot dynamics learning. The processing pipeline and data generation processes are presented in Sections III and IV. The network architectures and training

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methodologies are detailed in Section V, followed by an analysis of experimental results in Section VI. Finally, concluding remarks and future work are discussed in Section VII.

II. RELATED WORK

The literature on robot dynamics is evolving from classical system identification toward Physics-Informed Neural Networks. A primary example is Deep Lagrangian Networks (DeLAN) [1], which embeds the Euler-Lagrange equations into the architecture to ensure physical consistency and energy conservation. While DeLAN guarantees plausibility, it introduces higher computational complexity compared to standard models [2].

Conversely, Feed-Forward Neural Networks (FFNNs) are favored for their universal approximation capabilities but often function as "black boxes" that may violate physical principles [3]. Our work investigates if a trigonometric prior (using $\sin(q), \cos(q)$ as inputs) can bridge the performance gap by facilitating the learning of periodic gravity and Coriolis terms.

Finally, we address the limitations of the standard Mean Squared Error (MSE) in multi-joint manipulators. Because proximal joints exhibit significantly higher torque magnitudes than distal ones, MSE often ignores the dynamics of smaller joints. We implement a Normalized MSE (nMSE) to ensure balanced learning across all degrees of freedom.

Contribution Recap: This report extends existing benchmarks by: 1) evaluating input feature expansion in FFNNs, and 2) analyzing the role of nMSE in balancing multi-joint learning against DeLAN baselines.

III. PROCESSING PIPELINE

The proposed pipeline converts raw robotic state trajectories into joint torque predictions through a structured three-stage process: feature engineering, model configuration, and normalized evaluation.

A. Feature Engineering and Preprocessing

To investigate the impact of physical inductive biases on learning efficiency, we implement two distinct input strategies for the state vector:

- **Standard Input:** The network receives the raw kinematic state $\mathbf{x} = [q, \dot{q}, \ddot{q}]$.
- **Trigonometric Expansion:** The joint positions q are transformed into $[\sin(q), \cos(q)]$. This expansion aims to "de-burden" the network from approximating the periodic non-linearities inherent in Lagrangian dynamics, effectively providing a structural prior.

All features are subsequently standardized to zero mean and unit variance to ensure numerical stability during backpropagation.

B. Architectural Variations and Baseline Comparison

The study evaluates the trade-off between model depth and generalization by comparing:

- 1) **FFNN Depth:** We contrast a *Shallow* configuration against a *Deep* architecture to assess how increased non-linear capacity affects the mapping of complex inter-joint couplings.
- 2) **DeLAN Baseline:** The Feed-Forward models are benchmarked against the *Deep Lagrangian Network* (DeLAN), which serves as the physics-informed reference for robot dynamics.

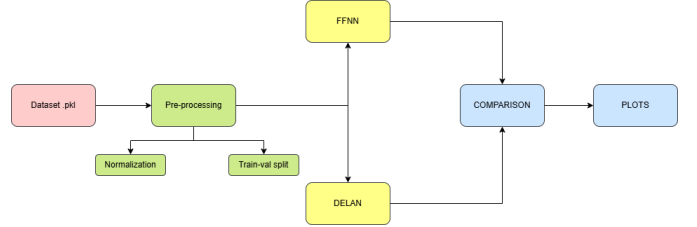


Fig. 1: Schematic representation of the proposed methodology

C. Training and Evaluation Metric

Models are optimized using the Adam optimizer. To ensure that joints with lower torque magnitudes are not neglected during training, we utilize the Normalized Mean Squared Error (nMSE) as the primary loss function and evaluation metric, obtained by computing MSE on normalized data:

$$nMSE = \frac{\mathbb{E}[(\tau - \hat{\tau})^2]}{\sigma^2(\tau)} \quad (1)$$

This has the effect of scaling the error of each joint relative to its own variance, allowing for a fair performance assessment across all Degrees of Freedom.

IV. SIGNALS AND FEATURES

A. Measurement Setup and Data Formatting

The dataset comprises simulated kinematic and dynamic trajectories of a Franka Emika PANDA robot arm. The robot's dynamics were formulated using modified Denavit-Hartenberg parameters, with the active degrees of freedom varying between 2 and 7 to represent different configuration complexities. Input trajectories were generated using a sum-of-sinusoids approach with a fundamental pulsation of 0.02 rad/s. The kinematic states and corresponding dynamics were sampled at a constant rate with a sampling time of 0.01 s. The resulting data is formatted as discrete multivariate time series, where the input at any given time step is represented by fixed-size vectors containing the robot's kinematic and dynamic state.

B. Signal Pre-processing and Feasibility

To ensure the generated signals closely mirror physical reality, the nominal dynamic parameters (such as link masses, centers of mass, and inertia tensors) were subjected to perturbation to simulate parameter uncertainty. During the trajectory generation phase, the signals were strictly constrained. A continuous validation check was applied to ensure that the generated joint positions q , velocities $\dot{q}$, and accelerations

$\ddot{q}$ strictly adhered to the physical operational limits of the PANDA robot. To simulate realistic sensor noise and imperfect measurements, the ground-truth joint torques were computed using the inverse dynamics model and subsequently corrupted with additive Gaussian noise with a standard deviation of 0.01.

C. Feature Extraction and Representation

The primary feature vectors extracted at each time step consist of the robot's fundamental kinematic variables: joint positions $q \in \mathbb{R}^n$, joint velocities $\dot{q} \in \mathbb{R}^n$, joint accelerations $\ddot{q} \in \mathbb{R}^n$. Alongside these raw kinematic inputs, the simulation explicitly calculates a suite of physics-informed features that fully describe the rigid-body dynamics of the system. These include the joint space inertia matrix $M \in \mathbb{R}^{n \times n}$, potential energy U , gravity torque vector $g \in \mathbb{R}^n$, and Coriolis/centrifugal torque vector $c \in \mathbb{R}^n$. Because the data is processed as continuous time series, segmentation can be applied dynamically during training depending on the sequence length requirements of the chosen network architecture. The primary target signal to be predicted by the model is the total measured joint torque $\tau \in \mathbb{R}^n$.

D. Dataset Splitting

To ensure rigorous model evaluation and prevent data leakage, the generated dataset was partitioned at the trajectory level using uniquely assigned random seeds. The final dataset is split into 60 independent trajectories for training, 21 for validation, and 50 for testing.

V. LEARNING FRAMEWORK

This section details the learning algorithms and architectures used to model inverse dynamics. The goal is to learn the mapping from joint kinematic states to actuation torques, comparing standard black-box approximators against physics-informed models.

A. Data Representation and Feature Engineering

Let n denote the number of Degrees of Freedom (nDOF) of the robotic manipulator. The standard input to the learning framework is the state vector $x_t \in \mathbb{R}^{3n}$ at time t , defined as:

$$x_t = [q_t^\top, \dot{q}_t^\top, \ddot{q}_t^\top]^\top, \quad (2)$$

where $q_t, \dot{q}_t, \ddot{q}_t \in \mathbb{R}^n$ represent the joint positions, velocities, and accelerations, respectively. The target output is the vector of joint torques $\tau_t \in \mathbb{R}^n$.

To facilitate learning in black-box architectures, we leverage the periodic nature of revolute joints by introducing a trigonometric expansion of the state space. This incorporates the system's inherent physics into the input features, relieving the network from learning these non-linear mappings from scratch. The modified input vector $\tilde{x}_t \in \mathbb{R}^{4n}$ is formulated as:

$$\tilde{x}_t = [\sin(q_t)^\top, \cos(q_t)^\top, \dot{q}_t^\top, \ddot{q}_t^\top]^\top. \quad (3)$$

Furthermore, the scale of the target torques varies significantly across the manipulator; base joints typically experience

much larger torques than the final joints. To ensure scale-independent learning and stable gradient descent, both inputs and outputs are standardized to zero mean and unit variance. Given a dataset $\mathcal{D}$, the normalized torque $\hat{\tau}$ is computed as:

$$\hat{\tau} = \left(\frac{\tau - \mu_\tau}{\sigma_\tau} \right), \quad (4)$$

where μ_τ and σ_τ are the empirical mean and standard deviation vectors of the torques in the training set.

B. Black-Box Architectures: Feed-Forward Neural Networks

Our baseline approach relies on standard Feed-Forward Neural Networks. Since these models are universal approximators, they do not encode any prior knowledge about rigid-body dynamics besides the engineered features.

We investigate the capacity of the model by comparing two structural paradigms:

- **Shallow Networks:** Architectures characterized by a limited number of hidden layers (typically 2 or 3). These networks are computationally lightweight but may struggle to capture the complex inter-joint Coriolis and centrifugal couplings.
- **Deep Networks:** Architectures employing a higher number of hidden layers (e.g., 5 to 7 layers) with widths ranging from 128 to 256 neurons. Deep architectures hierarchically compose non-linearities, theoretically providing a better functional approximation for high-DOF systems.

All hidden layers employ the Rectified Linear Unit (ReLU) activation function, and the networks are optimized using the Adam optimizer to find the parameter set θ^* that minimizes the empirical risk.

C. Physics-Informed Baseline: Deep Lagrangian Networks (DeLaN)

To counteract the limitations of pure black-box models, we benchmark our FFNNs against Deep Lagrangian Networks (DeLaN). Instead of directly outputting τ , DeLaN learns the fundamental components of the Euler-Lagrange equation: the mass matrix $M(q)$, the Coriolis/centrifugal forces $c(q, \dot{q})$, and the gravity vector $g(q)$.

The network is constrained to output a lower-triangular matrix $L(q)$ and a diagonal matrix $D(q) > 0$, ensuring that the reconstructed mass matrix $M(q) = L(q)D(q)L(q)^\top$ is strictly positive-definite. The torques are then derived analytically through automatic differentiation:

$$\tau = M(q)\ddot{q} + c(q, \dot{q}) + g(q). \quad (5)$$

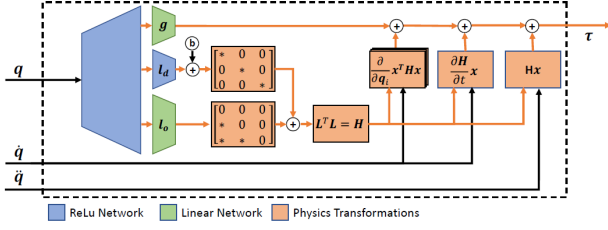


Fig. 2: The computational graph of the DeLaN

D. Evaluation Strategy and Metrics

To properly compare the trained models, the evaluation strategy extends beyond the standard Mean Squared Error (MSE), defined as $MSE = \frac{1}{N} \sum_{i=1}^N (\tau_i - \hat{\tau}_i)^2$.

As previously discussed, using MSE favors the minimization of error on base joints (which have large torque values) while neglecting the dynamics of distal joints. To evaluate performance equitably across all nDOF, we rely heavily on the normalized Mean Squared Error (nMSE), calculated independently for each joint $j \in \{1, \dots, n\}$:

$$nMSE_j = \frac{\frac{1}{N} \sum_{i=1}^N (\tau_{i,j} - \hat{\tau}_{i,j})^2}{\sigma_{\tau_j}^2}, \quad (6)$$

where $\sigma_{\tau_j}^2$ is the variance of the j -th joint torque in the test set. An aggregate nMSE is also computed to provide a single metric of global performance.

Finally, the models are evaluated qualitatively through trajectory tracking plots of predicted versus actual torques, and quantitatively on an Out-Of-Distribution (OOD) dataset. The OOD evaluation is critical to assess how severely the black-box FFNNs degrade compared to the physically structured DeLaN when subjected to unseen dynamic regimes. The results of these comparative analyses are detailed in Section VI.

VI. RESULTS

This section presents and discusses the experimental results, comparing the performance of the proposed architectures and loss functions in identifying the manipulator's inverse dynamics.

A. Comparison between network training methods

In this subsection, the performances of two FFNN trained with the nMSE and MSE loss are evaluated on a set of test trajectories, to assess the generalization capabilities in estimating the torque τ . Experimental data confirms that the network trained with MSE tends to prioritize joints with larger torque amplitudes, often resulting in suboptimal precision for joints characterized by smaller dynamic ranges. On the other hand, the network optimized via nMSE demonstrates superior performance when evaluated through the same metric, benefiting from the mathematical consistency between the training objective and the validation criteria. Notably, even when considering the worst-case scenario, represented by **Joint 2**, which exhibits the most challenging dynamics with a torque range of ± 20 Nm, the nMSE-trained model maintains a

MSE significantly below 1% of the full-scale excursion. This result validates that the nMSE formulation effectively balances the learning process across all Degrees of Freedom, preventing the optimization from disproportionately focusing on high-amplitude signals and ensuring high precision in the inverse dynamics estimation across the entire robot configuration.

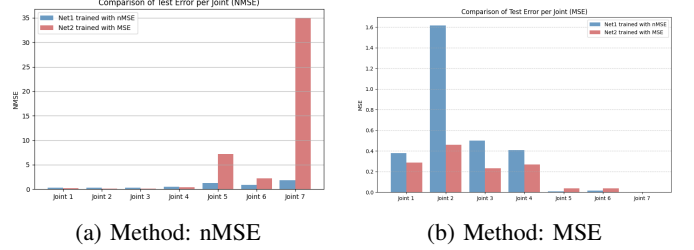


Fig. 3: Comparison between network training methods: (a) training with normalized loss, (b) training with standard loss.

B. Physics-Inspired Feature Engineering: Sin/Cos Mapping Performance

The inverse dynamics of manipulators with revolute joints are inherently periodic. To provide a structural inductive bias, we expand the joint positions q into a trigonometric state vector $[\sin(q), \cos(q)]$. This mapping removes the need for the network to approximate periodic nonlinearities, such as gravity and Coriolis terms from scratch.

Experimental results (Fig. 4) demonstrate that **Net2** (with sin/cos mapping) achieves enhanced precision in the robot's most critical degrees of freedom:

- **Joints 2 and 3:** These proximal joints, which support the highest gravitational loads, the plot show a significant reduction in MSE compared to the baseline. This confirms that the trigonometric prior effectively models gravity-dependent torques.
- **Joint 4:** The mapping maintains high accuracy, facilitating the transition between load-bearing proximal dynamics and motion-centric distal couplings.
- **Trade-offs:** While the structural prior excels at modeling periodicity, the increased error in the final joints suggests that complex high-order Coriolis and centrifugal effects may still require the increased representational capacity of deeper architectures.

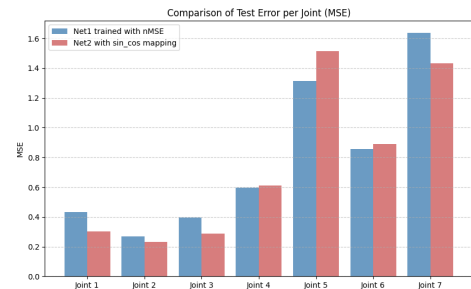


Fig. 4: Comparison between a baseline network trained with nMSE and the network trained on Sin/Cos mapped data

In summary, the $\sin/\cos$ expansion significantly improves regression accuracy in the primary joints (2, 3, and 4) by aligning the network's input space with the underlying Lagrangian physics.

C. Comparison between shallow and deep FFNN

Empirical evaluation of **FFNNs** for inverse dynamics demonstrates that increasing architectural depth improves global predictive capacity. Specifically, transitioning from a shallow baseline (2 layers: 128, 256 units) to a deep architecture (5 layers: 128 to 128 units) reduced the average **nMSE** from 0.766 to 0.626.

This gain is localized at the **distal joints (5–7)**, where the deep network better approximates high-order non-linear dynamic couplings. Conversely, for **proximal joints** governed by gravity and primary inertia, increased depth offers negligible advantages and occasional marginal degradation due to localized overfitting.

Despite these improvements, substantial errors at the end-effector ($\text{nMSE} > 1.0$) highlight the limitations of purely data-driven models. Standard black-box architectures lack the inductive biases to enforce mechanical priors, such as energy conservation or a positive-definite mass matrix. Consequently, increasing depth alone is insufficient for robust generalization, necessitating **physics-informed constraints** for dynamically consistent torque tracking.

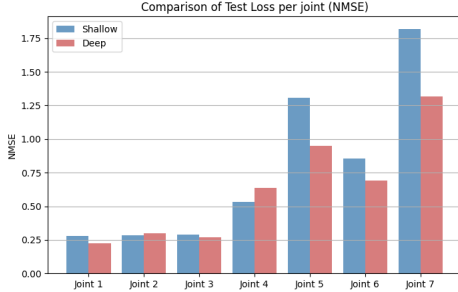


Fig. 5: Comparison between shallow and deep neural network

D. Generalization Limitations and Out-of-Distribution Performance

To evaluate robustness, FFNNs were tested on an out-of-distribution (OOD) dataset featuring unseen kinematic states ($q, \dot{q}, \ddot{q}$). Results indicate a severe degradation in predictive performance, suggesting that the networks overfit training manifolds rather than learning underlying physical laws.

Paradoxically, the deep architecture's expanded capacity becomes a liability OOD, yielding a higher average MSE (5.29) than the shallow baseline (4.35). Generalization fails catastrophically at proximal joints (2, 3, and 4), which bear the heaviest loads; for Joint 2, the deep network's MSE exceeds 20.0, significantly higher than the shallow network's 14.6. This confirms that increased depth leads to the memorization of spurious correlations, resulting in erroneous extrapolations for novel kinematic configurations.

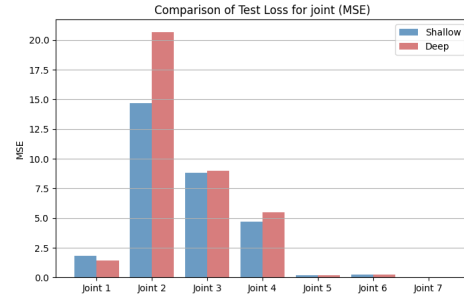
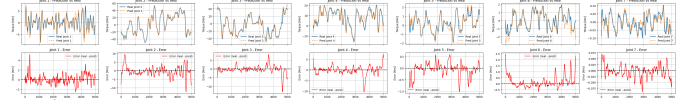
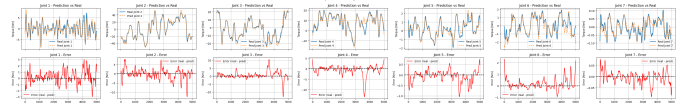


Fig. 6: Comparison between shallow and deep FFNN with data out of distribution

The temporal trajectory evaluations further corroborate these structural limitations. Despite capturing low-frequency trends, both networks fail during highly dynamic transients, exhibiting severe overshoots, phase lags, and unphysical oscillations during high accelerations or rapid velocity reversals. The deep network, in particular, generates larger spurious torque predictions at these critical junctures. These OOD testing results empirically demonstrate a fundamental vulnerability of standard black-box architectures in robotic control: without the explicit embedding of mechanical laws, FFNNs act as localized curve-fitters rather than generalized inverse dynamics models. They cannot reliably extrapolate physical relationships—such as the quadratic dependence of Coriolis forces on velocity or the trigonometric dependence of gravity on position—outside their immediate training domain. This profound lack of generalization further motivates the necessity of physics-informed frameworks, where the hypothesis space is strictly bounded by analytical mechanics to guarantee stable and predictable behavior across the entire operational workspace.



(a) Predicted vs. actual joint torques and associated tracking errors for the shallow network



(b) Predicted vs. actual joint torques and associated tracking errors for the deep network

Fig. 7: Comparative evaluation of temporal torque tracking on the out-of-distribution dataset

E. DeLAN performances and comparison with FFNN

The **DeLAN** was evaluated as a physics-informed alternative to the purely data-driven **FFNN**. Despite its ability to enforce energy conservation and physical consistency, the model exhibited suboptimal performance in this implementation due to the following factors:

- **Loss Function Dependency:** DeLAN utilized a standard **MSE loss**, which, unlike the proposed **nMSE**, focuses too much on high-torque proximal joints while neglecting the dynamics of distal links.
- **Computational Constraints:** Due to limited hardware resources, the network was trained for only **50 epochs**. Given the highly non-linear nature of the Loss function, this duration was insufficient for full convergence. Standard benchmarks typically require upwards of **1000 epochs** to realize the benefits of the structural prior.
- **Out-of-Distribution (OOD) Performance:** The model showed high error in OOD regimes. However, the failure mode was consistent with the MSE-driven bias observed in training, suggesting that with extended training and loss normalization, the model's inherent physical robustness could still be realized.

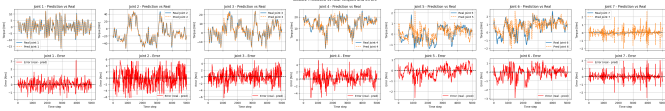
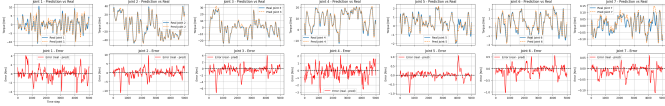
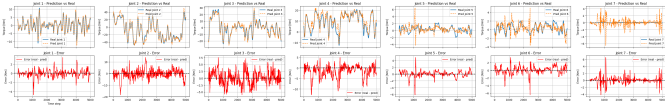


Fig. 8: DeLAN torque tracking on the test dataset



(a) Shallow FFNN torque tracking on the OOD dataset



(b) DeLAN torque tracking on the OOD dataset

Fig. 9: Comparative evaluation of temporal torque tracking on the out-of-distribution dataset

In conclusion, while the FFNN with nMSE proved more effective under restricted training budgets, the DeLAN remains a promising candidate for high-fidelity tracking if the optimization bottleneck is addressed through increased epochs and balanced metrics.

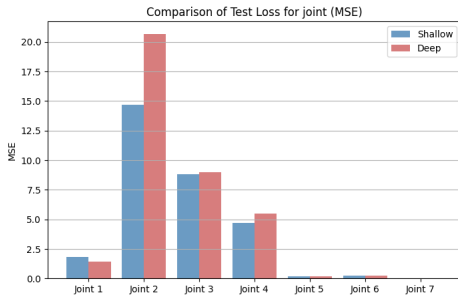


Fig. 10: Error comparison between DeLAN and the shallow FFNN on the OOD dataset

VII. CONCLUDING REMARKS

In this study, we evaluated the performance of data-driven FFNNs and physics-informed DeLAN architectures in approximating the inverse dynamics of a 7-DOF manipulator. We demonstrated that the adoption of a per-joint nMSE loss function is essential for balancing learning across proximal and distal joints. Furthermore, providing a trigonometric structural prior ($\sin/\cos$) effectively "de-burden" the network from learning periodic gravitational components from scratch, significantly improving accuracy in joints 2, 3, and 4.

A. Observations and Future Work

A key finding is the trade-off between architectural simplicity and physical consistency. While deeper FFNNs improve distal tracking, they lack the energy-conservation guarantees of DeLAN. However, DeLAN's reliance on standard MSE and its extreme computational demand make it difficult to optimize under restricted budgets. Future work should focus on integrating nMSE directly into the Lagrangian framework and conducting extensive training (at least 1000 epochs) to allow the complex $M(q)$ and $c(q, \dot{q})$ terms to converge fully.

B. Learning Experience and Difficulties

This project provided a deep dive into the practicalities of robotic simulation and multivariate time-series processing. We learned that:

- **Resource Management:** Physics-informed models involving higher-order derivatives require significant computational power and time.
- **Data Engineering:** We managed to practice with some robotics data.
- **Critical Challenges:** The primary difficulty was balancing the learning scales across joints. This taught us that the MSE loss function is often insufficient for complex mechanical systems.

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